

Asymmetric Manifold Embedding via Randers-UMAP

Method Overview

1 Randers Distance and the Asymmetric Graph

Given a point cloud $X = \{x_1, \dots, x_n\} \subset \mathbb{R}^m$ and a vector field $\omega : X \rightarrow \mathbb{R}^m$, the *Randers metric* at x is

$$F_x(v) = \|v\| + \langle \omega_x, v \rangle. \quad (1)$$

Both pipelines in this project ultimately produce an asymmetric matrix $D_{asym} \in (\mathbb{R}_{\geq 0} \cup \{\infty\})^{n \times n}$, interpreted as the directed geodesic distance on a weighted directed graph $G = (V, E, w)$, $V = \{1, \dots, n\}$:

$$D_{asym}(i, j) = \min_{\text{directed path } i \rightsquigarrow j \text{ in } G} \sum_{(p, q) \in \text{path}} w(p, q), \quad D_{asym}(i, j) = \infty \text{ if no such path exists.} \quad (2)$$

The two families below differ entirely in how the pair (E, w) is chosen – everything downstream of D_{asym} is shared (Section 7).

2 Two Constructions of D_{asym}

2.1 Generated

Edges are either the directed k -nearest-neighbour relation $E_{knn} = \{(i, j) : x_j \in \text{kNN}_k(x_i)\}$ (asymmetric: $j \in \text{kNN}_k(i)$ does not imply $i \in \text{kNN}_k(j)$), or the symmetric ε -ball relation $E_\varepsilon = \{(i, j) : \|x_i - x_j\| < \varepsilon\}$. Edge weight is the discretized Randers length:

$$w(i, j) = \|x_i - x_j\| + \langle \omega_i, x_j - x_i \rangle, \quad (i, j) \in E. \quad (3)$$

D_{asym} is then the full shortest-path closure of (V, E, w) . Since G is (enforced to be) connected, every pair has a finite multi-hop geodesic, so D_{asym} is **dense**.

2.2 Calculated

For each i , let $N_k(i)$ be its k nearest neighbours under $\|\cdot\|$ – IsUMap’s own local, *radial* neighbourhood (distances measured only from i outward, not merged with any other point’s neighbourhood):

$$D_{asym}(i, j) = \begin{cases} \|x_i - x_j\| & j \in N_k(i) \\ \infty & \text{otherwise.} \end{cases} \quad (4)$$

No shortest-path closure and no fuzzy union of overlapping neighbourhoods is taken here: D_{asym} keeps exactly k finite entries per row and is therefore genuinely **sparse**. (IsUMap’s reference algorithm would, at this point, additionally form a T-conorm fuzzy union of the local neighbourhoods and *then* take a shortest-path closure before use; this project uses the un-merged, un-closed neighbourhood relation directly – verified empirically that a real edge (i, j) frequently has no matching edge (j, i) , which is exactly the *existence asymmetry* the drift extraction in Section 6 is built to exploit. The discarded merge/closure step was, until recently, still being computed and thrown away every run; this has since been eliminated as pure wasted computation, with no change to D_{asym} ’s value.)

2.3 Consequence

Generated’s asymmetry has two superposed sources: the field ω , and (under the k -NN edge rule) the directed existence structure of E_{knn} itself. Calculated’s asymmetry has exactly one source: the directed existence structure of $N_k(\cdot)$ – there is no field. Both graphs, however, now exhibit the same qualitative phenomenon: $(i, j) \in E \not\Rightarrow (j, i) \in E$.

3 D_{geo} : a Field-Free Companion Matrix

Needed wherever the force field (Section 7) is asked to evaluate an ”Euclidean” (non-Randers) term whose *weight* should not itself carry drift information.

- **Generated:** D_{geo} is the shortest-path closure of the same graph with $\omega \equiv 0$, i.e. $w(i, j) = \|x_i - x_j\|$ – an independent closure.
- **Calculated:** derived algebraically from D_{asym} itself, no second closure:

$$D_{geo}(i, j) = \begin{cases} \frac{1}{2}(D_{asym}(i, j) + D_{asym}(j, i)) & \text{both finite} \\ D_{asym}(i, j) \text{ or } D_{asym}(j, i) & \text{exactly one finite} \end{cases} \quad (5)$$

(An unconditional average, requiring both directions finite, would leave some rows sparser in D_{geo} than in D_{asym} itself – the case-split above guarantees D_{geo} is never sparser than D_{asym} .)

4 Initialization

Classical (Torgerson) MDS maps a dense, symmetric dissimilarity matrix Δ to coordinates via double centering, followed by a rank- d eigendecomposition of B . Both families feed it a matrix built the same way as D_{asym} itself – a shortest-path closure of a k -NN/ ε graph (Section 2) – just symmetrized and/or taken over a different point set:

4.1 Generated

Embedding X directly under ω would require carrying a directional weight through MDS itself, which classical MDS cannot represent (its double-centering step assumes a symmetric Δ). Instead ω is baked into a second copy of the data *geometrically*: define the virtual points $x'_i = x_i + \omega_i$ and the augmented set $\tilde{X} = X \cup \{x'_i\}_{i=1}^n$, $|\tilde{X}| = 2n$. Then Δ is the (undirected) shortest-path closure of the *same* k -NN/ ε graph rule as Section 2.1, but with the plain Euclidean weight $w(p, q) = \|p - q\|$ – no Randers term is needed here, since ω is already encoded in where x'_i sits relative to x_i , not in an edge weight. Classical MDS on Δ places $\tilde{Y} \in \mathbb{R}^{2n \times d}$; splitting it by the original partition, $\tilde{Y} = (Y_{real}, Y_{virtual})$, recovers each point and its shifted twin *in the same coordinate frame* – their difference is exactly the frozen drift of Section 5.

4.2 Calculated

Δ is the *directed* shortest-path closure of D_{asym} itself – i.e. D_{asym} ’s own sparse, radial k -NN entries are used as graph edges and Dijkstra is re-run on them to reach every pair, filling any pair left unreachable (different strongly-connected components) with the graph’s own finite diameter, so the squaring/centering step in B stays well-defined. This closure exists only to seed Y_{init} and is discarded immediately afterward: it never touches the sparse D_{asym} used for training.

5 The Drift Field b

Two schemes populate the per-node drift vector $b_i \in \mathbb{R}^d$ that encodes directionality into the embedding.

Frozen (Generated’s default).

$$b_i = (Y_{\text{virtual}})_i - (Y_{\text{real}})_i, \quad \text{fixed for the entire optimization.} \quad (6)$$

Live (Calculated’s default). A fixed per-pair asymmetry signal is extracted once from D_{asym} (Section 6 gives N), then, at *every* step t , using the current embedding $y^{(t)}$:

$$b_i^{(t)} = \frac{1}{k} \sum_{j \in N_k(i)} N(i, j) e_{ij}(y^{(t)}), \quad e_{ij}(y) = \frac{y_j - y_i}{\|y_j - y_i\|}, \quad (7)$$

clipped to $\|b_i\| \leq 1 - \delta$. Direction tracks the live layout; only N is fixed.

Frozen-from-topology (new). The same N and the same formula for b_i above, but evaluated once at $y = Y_{\text{init}}$ and held fixed thereafter – structurally identical to the Generated scheme, but with N (data asymmetry) playing the role ω (a known field) plays there. This isolates *freezing* from *where the asymmetry came from*: the same graph D_{asym} can now be trained under either a live or a frozen drift.

6 Existence-Asymmetry Correction in N

D_{asym} need not be dense: a pair can have $D(i, j) < \infty$, $D(j, i) = \infty$. Naively,

$$N(i, j) = \frac{D(i, j) - D(j, i)}{D(i, j) + D(j, i) + \varepsilon} \quad (8)$$

sends such a pair to $\frac{\infty - \infty}{\infty}$, undefined – previously mapped to $N(i, j) = 0$, treating a one-directional pair as “no signal”, even though a missing reverse relation is arguably the *strongest* asymmetry signal available. Fixed by substituting the graph’s own diameter for the missing side before forming N :

$$\text{diam} = \max_{D(p, q) < \infty} D(p, q), \quad \text{diam}^+ = \text{diam} \cdot (1 + 10^{-6}) \quad (9)$$

$$D^{\text{fill}}(i, j) = \begin{cases} D(i, j) & D(i, j) < \infty \\ \text{diam}^+ & \text{otherwise} \end{cases} \quad (10)$$

$$N(i, j) = \frac{D^{\text{fill}}(i, j) - D^{\text{fill}}(j, i)}{D^{\text{fill}}(i, j) + D^{\text{fill}}(j, i) + \varepsilon} \in [-1, 1]. \quad (11)$$

Pairs with *no* information in either direction ($D(i, j) = D(j, i) = \infty$, e.g. disconnected components) still get $N(i, j) = 0$. Using $\text{diam}^+ > \text{diam}$ strictly (rather than $=$) prevents a pair whose known direction happens to equal the graph’s own diameter from cancelling to exactly 0.

7 The Shared Embedding Functional

From D_{asym} onward (and D_{geo} , $b^{(0)}$, Y_{init} where relevant), both families are optimized by *identical* machinery.

7.1 High-dimensional membership

Following UMAP’s own fuzzy-simplicial-set construction (McInnes, Healy & Melville, arXiv:1802.03426, §3.1): with $\rho_i = \min_j D_{\text{asym}}(i, j)$ and σ_i calibrated so $\sum_j \exp(-\max(0, D_{\text{asym}}(i, j) - \rho_i)/\sigma_i) = \log_2 k$,

$$\mu(i, j) = \exp\left(-\frac{\max(0, D_{\text{asym}}(i, j) - \rho_i)}{\sigma_i}\right), \quad (12)$$

computed once (D_{asym} is static); μ is directed, $\mu(i, j) \neq \mu(j, i)$ in general.

7.2 Randers substitution

UMAP’s Euclidean distance/direction terms are replaced by their Randers-perturbed counterparts, evaluated at the *live* embedding y :

$$\rho(i \rightarrow j) = \|y_i - y_j\| + b_i \cdot (y_j - y_i), \quad g(i, j) = e_{ij}(y) + b_i. \quad (13)$$

$b \equiv 0$ recovers vanilla UMAP exactly ($\rho = \|y_i - y_j\|$, $g = e_{ij}$).

7.3 Force law

Two interchangeable choices for how (ρ, g) produce a force:

UMAP’s own curve (paper §3.2), with (a, b_*) fit to a target (min_dist, spread):

$$\alpha(\rho) = \frac{2ab_* \rho^{2b_*-1}}{1 + a\rho^{2b_*}}, \quad \beta(\rho) = \frac{2b_* \rho}{\rho^2 (1 + a\rho^{2b_*})} \quad (14)$$

$$F(i, j) = \mu_{\text{attr}}(i, j) \alpha(\rho_{\text{attr}}) g_{\text{attr}} - \lambda (1 - \mu_{\text{rep}}(i, j)) \beta(\rho_{\text{rep}}) g_{\text{rep}}, \quad (15)$$

where $\mu_{\text{attr}} = \mu(i, j)$ or a field-free companion built from D_{geo} (Section 4) if the attractive/repulsive term is requested Euclidean rather than Randers, and λ rescales the dense repulsion sum to the same expected magnitude as stochastic negative sampling.

Force-directed social gravity (Bannister, Eppstein, Goodrich & Trott, GD 2012, arXiv:1209.0748; current default), with natural edge length $k_{\text{nat}} = 1/\sqrt{n}$ by default:

$$F_a(i, j) = \frac{\rho(i, j)}{k_{\text{nat}}} g(i, j), \quad (i, j) \in E \cup E^\top \text{ only} \quad (16)$$

$$F_r(i, j) = \frac{k_{\text{nat}}^2}{\rho(i, j)^2} g(i, j), \quad \text{all pairs, unconditional.} \quad (17)$$

7.4 Additional force terms

All independent, all additive on top of $F_i = \sum_j F(i, j)$:

$$\text{gravity:} \quad F_i += \gamma M_i w_i b_i, \quad w_i = \exp\left(-\frac{\max(0, \|b_i\| - \rho_i^{\text{loc}})}{\sigma_i^{\text{loc}}}\right) \quad (18)$$

$$\text{virtual neighbour:} \quad F_i += \alpha(\|b_i\|) b_i, \quad (\text{unconditional, } \mu_{\text{virtual}} \equiv 1) \quad (19)$$

$$\text{negative sampling:} \quad F_r(i, j) \rightarrow \text{Monte-Carlo estimate over } S \text{ random } j, \quad \text{rescaled to the same expectation.} \quad (20)$$

where $\rho_i^{\text{loc}}, \sigma_i^{\text{loc}}$ are i ’s own live local-neighbourhood scale in y (gravity’s plausibility weighting), and M_i is a per-node mass (uniform by default).

7.5 Optimization

An optional ramp $s(t) \in [0, 1]$ (held at 0 for the first 30% of steps, linear over the next 40%, held at 1 for the last 30%) scales $b^{(t)} \leftarrow s(t) b^{(t)}$ before every force evaluation. Updates use $y += \eta(t) \text{clip}(F, \pm c)$ with a linearly decaying step size $\eta(t) = \eta_0(1 - t/T)$.

8 Experimental Testbeds

- **Swiss roll, mammoth, sphere (radial/tangential):** synthetic X with a hand-designed ground-truth ω (Generated) run in parallel with IsUMap-derived asymmetry on the identical X (Calculated) – recovered b can be checked against the known ω .
- **MNIST (raw/PCA), Breast Cancer, scRNA:** real high-dimensional data, Calculated family only – no ground-truth field exists; asymmetry is whatever $N_k(\cdot)$ ’s directed structure exhibits.

9 Summary

	Generated	Calculated
Asymmetry source	ω (+ directed k -NN existence)	directed $N_k(\cdot)$ existence only
D_{asym}	dense (shortest-path closure)	sparse (k entries/row, no closure)
D_{geo}	independent field-free closure	algebraic, from D_{asym}
Y_{init} from	closure of $X \cup (X + \omega)$	throwaway closure of D_{asym}
Default b	frozen, from ω	live, from N
Also available	–	frozen, from N (new)

From the shared embedding functional (Section 7) onward, both families are optimized by exactly the same objective: only the graph (E, w) and, optionally, the frozen-vs-live choice for b differ upstream of it.